CHAPTER 6 · MOCK TEST

Distribute & Multiply — Full Test

Mixed questions from all three concepts. Section A is interactive; Sections B and C are written. Time: 45 minutes · Total: 28 marks. Answers on the separate key.

Objective (1 mark each)

Q1 · MCQ **[1]**

$a(b + c)$ equals:

(a) $ab + c$ (b) $ab + ac$ (c) $a + bc$ (d) abc

Q2 · MCQ **[1]**

$(a + b)^2$ equals:

(a) $a^2 + b^2$ (b) $a^2 + 2ab + b^2$ (c) $a^2 - 2ab + b^2$ (d) $2ab$

Q3 · MCQ **[1]**

$(a + b)(a - b)$ equals:

(a) $a^2 + b^2$ (b) $a^2 - b^2$ (c) $a^2 - 2ab + b^2$ (d) $a^2 + 2ab + b^2$

Q4 · MCQ **[1]**

$-3p(-5p + 2q)$ equals:

(a) $15p^2 - 6pq$ (b) $-15p^2 + 6pq$ (c) $-3p + 5p - 2q$ (d) $15p^2 + 6pq$

Q5 · FILL IN THE BLANK **[1]**

Using an identity, 99^2 = _______.

____

Q6 · TRUE / FALSE **[1]**

a^2b and ab^2 are like terms.

True False

Q7 · MCQ **[1]**

The correct expansion of $(5m + 6n)^2$ is:

(a) $25m^2 + 36n^2$ (b) $25m^2 + 60mn + 36n^2$ (c) $25m^2 - 60mn + 36n^2$ (d) $30mn$

Q8 · ASSERTION–REASON **[1]**

Assertion (A): $397 \times 403 = 159991$.
Reason (R): $(a + b)(a - b) = a^2 - b^2$.

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Short answer (2 marks each)

Q9 **[2]**

Expand $(3 + u)(v - 3)$.

Q10 **[2]**

Expand $(6x + 5)^2$.

Q11 **[2]**

Find 55^2 using a suitable identity.

Q12 **[2]**

Find and correct the mistake: $2(x - 1) + 3(x + 4) = 5x + 3$.

Long answer (3 marks each)

Q13 **[3]**

Expand $(10a + b)(10c + d)$.

Q14 **[3]**

Use identities to compute: (i) 46^2 (ii) 397×403.

Q15 **[3]**

Expand $(4y + 7)(y + 11z - 3)$.

Q16 **[3]**

Without fully multiplying, decide which is larger: 14×26 or 16×24. Justify using an identity.

Total: 28 marks · Check your work against the answer key.